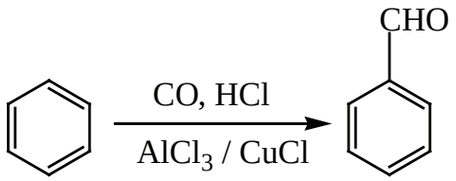
Oxidation of toluene by chromic oxide gives benzylidine diacetate which on hydrolysis gives benzaldehyde.

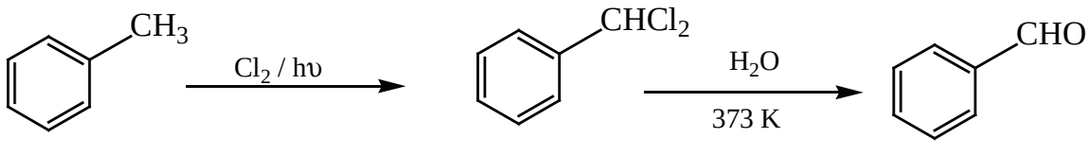
2) Gattermann – Koch reaction

This reaction is a variant of Friedel – Crafts acylation reaction. In this method, reaction of carbon monoxide and HCl generate an intermediate which reacts like formyl chloride.



3) Manufacture of benzaldehyde from toluene

Side chain chlorination of toluene gives benzal chloride, which on hydrolysis gives benzaldehyde.



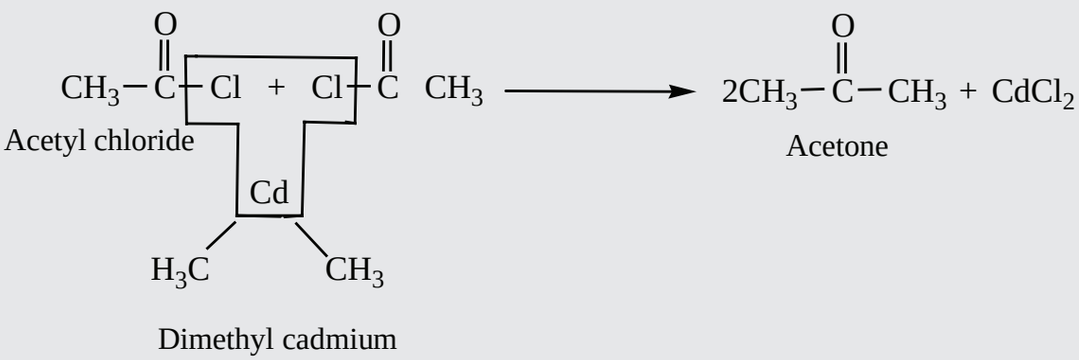
Toluene — Benzalchloride — Benzaldehyde

This is the commercial method for the manufacture of benzaldehye.

D) Preparation of ketones

1) **Ketones** can be prepared by the action of acid chloride with dialkyl cadmium.

Example



2) Preparation of phenyl ketones

Friedel – Crafts acylation

It is the best method for preparing alkyl aryl ketones or diaryl ketones. This reaction succeeds only with benzene and activated benzene derivatives.

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